



Antonio Desiderio

POSTDOCTORAL RESEARCHER

Turin, Italy

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🎓 Antonio Desiderio | ORCID

I build computational models of complex social and urban networks, and develop data pipelines and software to analyze human dynamics, mobility patterns, and financial market behaviors.

Employment

ISI Foundation

Turin, Italy

POSTDOCTORAL RESEARCH

Aug. 2025 - Currently

- I am a postdoctoral researcher in Urban Dynamics at the ISI Foundation, working with prof. Riccardo Di Clemente. I work on the COLINE (Complex Links of NEighbourhoods) DUT European Partnership project (F-DUT-2023-0037). More information can be found here: COLINE

Technical University of Denmark (DTU Compute, CogSys Section)

Copenhagen, Denmark

POSTDOCTORAL RESEARCH

Jan. 2024 - Aug. 2025

- I was a postdoctoral researcher in Urban Multimodal Mobility Modeling at the Technical University of Denmark in the Social Complexity Lab, led by Sune Lehmann and Laura Alessandretti. I have worked on the GEMINI (Greening European Mobility through cascading innovation INitatives) project funded by the European Union's Horizon Innovation Actions programme under Grant agreement No. 10110380, and lead the project from August 2024 to March 2025. More information can be found here: GEMINI

Bank of Italy

Rome, Italy

DATA SCIENTIST

Oct. 2019 - Apr. 2020

- I have built and analysed the Knowledge Graph of Italian Companies (KG-IC), which was employed during the COVID-19 crisis in Italy (Golden Power). KG-IC was constructed by employing web scraping techniques to extract information about companies, integrated with the Bank of Italy database and followed by a geocoding process, to uncover the interplay between local and non-local relationships among Italian businesses. More informations can be found here: ART, KG-COVID-19

Publications & Reviewers

Peer Reviewed

- A. Desiderio et al., "Mapping regional disparities in discounted grocery products" [doi:10.1038/s41538-026-00764-0]
- A. Desiderio et al., "Highly engaging events reveal semantic and temporal compression in online community discourse" [doi:10.1093/pnasnexus/pgaf056]
- A. Desiderio et al., "The dynamics of the Reddit collective action leading to the GameStop short squeeze" [doi:10.1038/s44260-025-00029-z]
- G. Palermo, A. Desiderio et al., "Spontaneous Opinion Swings in the Voter Model with Latency" [doi:10.1103/PhysRevE.110.024313]
- A. Desiderio et al., "Multiplex mobility network and metapopulation epidemic simulations of Italy based on open data" [doi:10.1088/2632-072X/ac9a29]
- A. Mancini, A. Desiderio et al., "Self-induced consensus of Reddit users to characterise the GameStop short squeeze" [doi:10.1038/s41598-022-17925-2]
- A. Desiderio et al., "First lattice calculation of radiative leptonic decay rates of pseudoscalar mesons" [doi:10.1103/PhysRevD.103.014502]

Pre-print

- Y. Liao, ..., and A. Desiderio, "Urban mobility enables deprivation bubble breaking in Indian and Mexican cities" (Under review on Nature Communications) [doi:arXiv.2603.29782]
- D. Cirulli, A. Desiderio et al., "Polarization and echo chambers in Reddit's political discourse" (Under review on npj Complexity) [doi:arXiv.2510.27467]
- A. Mancini, A. Desiderio et al., "The rise and fall of WallStreetBets: social roles and opinion leaders across the GameStop saga" (Under review on Scientific Reports) [doi:arXiv.2403.05876]

Reviewer

- Nature Portfolio: Nature Communications, Nature Communications Physics, Nature Cities.
- Physical Review Journals: Physical Review E, Physical Review Research.
- SpringerOpen Portfolio: Humanities and Social Sciences, EPJ Data Science, Applied Network Science.
- Elsevier: Global Challenges & Regional Science.
- IEEE: Transactions on Computational Social Systems.
- IOP: Journal of Statistical Mechanics: Theory and Experiment.
- World Scientific Publishing: Advances in Complex Systems.

Research Outreach

Research Outreach

- 2026, GovLab Report, “2024-2026, The Re-Use of Non-Traditional Data for Public Interest Purposes”, two scientific works featured in the report ([doi:10.1038/s41538-026-00764-0] and [doi:10.1093/pnasnexus/pgaf056]), [report thegovlab.org]
- 2025, Press Release on DTU News, “Changing gears: DTU helps Rudersdal embrace shared mobility”, [news.dtu.dk]
- 2025, Press Release on Northeastern News, “From politics to sports, Reddit conversations narrow and intensify before major events, new research finds”, [news.northeastern.edu]
- 2024, Press Release on phys.org, “Researchers show Reddit users caused the famous GameStop ‘short squeeze’”, [phys.org]
- 2024, External Collaborator of IPPr: Institute for Public Policy Research, UK’s export diversity over the last decade using Fitness and Economic Complexity, [report]
- 2023, DataBeers Rome, Public Talk
- 2022, Press Release on Nature Behind the Paper, “Did WallStreetBets increase GameStop popularity to drive the market?”, [link]

Projects, Grants & Visiting

Project

- 2026, Smart Bites: Decoding Denmark’s Grocery Patterns to reduce Food Waste, Salling Group and Technical University of Denmark, Co-PI with Sune Lehmann

Grant

- 2025, Travel Grant, Otto Moensted Foundation, Grant ID 25-81-1979 (2.000 DKK)
- 2024, Visiting Grant, SoBigData++ Transnational Access, The impact of urban environments on navigation strategies (5.000 EU)
- 2023, Visiting Grant, Danish Data Science Academy, Grant ID DDSA-V-2023-008 (15.000 DKK)

Visiting

- October-December 2024, ISTI Institute of CNR, Department of Computer Science University of Pisa and Scuola Normale Superiore (Pisa), Knowledge Discovery and Data Mining Laboratory, Mobility Team, Luca Pappalardo
- May-July 2023, DTU Compute - Technical University of Denmark (Copenhagen), Section for Cognitive Systems, Social Complexity Lab, Laura Alessandretti

Talks & Conferences

Invited Talks

- 2024, Pisa, ISTI Institute of CNR
- 2024, Oxford, University of Oxford, Network Seminar
- 2024, Copenhagen, IT University of Copenhagen
- 2023, Copenhagen, Technical University of Denmark

Program Committee

- HHA1, International Conference Series on Hybrid Human-Artificial Intelligence, Edition 2025, 2026
- CCS, Conference on Complex Systems, Edition 2024, 2025, 2026
- NETSCI, International School and Conference on Network Science, Edition 2024, 2025, 2026
- IC2S2, International Conference on Computational Social Science, Edition 2025, 2026
- CS2Italy, The Italian Conference on Computational Social Science, Edition 2026

Contributed Speaker

- 2026 CS2Italy, Turin, Italy, Main Conference, “How Spatial Proximity Affects Urban Activities Resilience: the Case of COVID-19”
- 2025 CCS, Siena, Italy, Main Conference, “How Spatial Proximity Affects Urban Activities Resilience: the Case of COVID-19”
- 2025 Complexity in Economics & Finance, FinEcoNets & Econophysics Colloquium StatPhys29 Satellite, Rome, Main Conference, “The Dynamics of The Reddit Collective Action Leading to the Gamestop Short Squeeze”
- 2024 CompleNet, Exeter, Main Conference, “Recurring Patterns in Online Social Media Interactions during Highly Engaging Events”
- 2024 CompleNet, Exeter, Main Conference, “How Spatial Proximity Affects Urban Activities Resilience: the Case of COVID-19”
- 2023 IC2S2, Copenhagen, Main Conference, “Self-induced consensus of Reddit users to characterise the GameStop short squeeze”
- 2023 IC2S2, Copenhagen, Main Conference, “Recurring patterns within Reddit communities during exogenous events”
- 2022 CCS, Palma de Mallorca, Computational Social Science Satellite, “Emergent similarities among Reddit communities during exogenous events”
- 2022 NETSCI, Shanghai (Online), Main Conference, “Self-induced consensus of Reddit users to characterise the GameStop short squeeze”

Chair, Committee, Organiser & Volunteer

- 2025, ISI Foundation, CS2Italy, Local Organizing Committee
- 2025, ISI Foundation, 2nd European Mobility Symposium, Web Chair
- 2024, National Research Council Pisa, European Mobility Symposium, Steering Committee
- 2024, Exeter, CompleNet, Session Biological and Economical Networks, Chair
- 2023, IMT Lucca, 1st Workshop “Social Network Dynamics and Impact on Financial Markets”, Organiser
- 2023, Copenhagen, 9th International Conference on Computational Social Science (IC2S2), Volunteer

References

References

- Dr Giulio Cimini, Associate Professor, University of Tor Vergata and Centro Fermi, Rome, Italy
giulio.cimini@roma2.infn.it
- Dr Riccardo Di Clemente, Associate Professor, Northeastern University, London, United Kingdom
riccardo.diclemente@nulondon.ac.uk
- Dr Laura Alessandretti, Associate Professor, Technical University of Denmark, Copenhagen, Denmark
lauale@dtu.dk

Education

Ph.D. & Doctor Europaeus in Physics, University of Rome Tor Vergata and Enrico Fermi Research Center

Rome, Italy

PHYSICS

Oct. 2020 - Oct 2023

- Final Grade: excellent quality cum laude
- Thesis Title: Complex dynamics in social systems during recent exogenous and endogenous events
- Topics: Complex Networks, Epidemic Spreading, Opinion Dynamics, Online Engagement
- Supervisors: Professor Giulio Cimini
- Examiners: Professor Nicola Perra, Professor Fabiana Zollo

Master Degree, University of Rome Tor Vergata

Rome, Italy

THEORETICAL PHYSICS

Dec. 2016 - May 2019

- Final Grade: 110/110 cum laude
- Thesis Title: Lattice calculation of the radiative leptonic decay rates of pseudoscalar mesons
- Topics: High Performance Computing, Quantum Field Theory, Big Data Analysis
- Supervisors: Professor Nazario Tantalo, Dr Martin Hansen

Bachelor Degree, University of Rome Tor Vergata

Rome, Italy

PHYSICS

Oct. 2013 - Dec. 2016

- Final Grade: 110/110 cum laude
- Thesis Title: Formal theory of Scattering: the case of Coulomb potential
- Topics: Quantum Field Theory
- Supervisors: Professor Nazario Tantalo

Teaching & Supervision

Teaching

- 2026, Guest Lecturer, Northeastern University of Boston, Network Science Course (Master Degree), CNET5341

Master Thesis Co-Supervisor

- Polytechnic University of Milan, Master Degree Mathematical Engineering - Quantitative Finance: Margherita Bencini. Co-supervision with Prof. Paolo Barucca, Daniele Marazzina.
- University of Tor Vergata, Master Degree Physics: Anna Mancini, Gioacchino Mauri, Giulia Di Palma, Daniele Cirulli. Co-supervision with Prof. Giulio Cimini, Prof. Riccardo Di Clemente, Prof. Mauro Sbragaglia, Dr. Francesca Pelusi, Dr. Fabio Saracco.
- University of Exeter, MSc Data Science with Artificial Intelligence: Nawaf Alaqeili, Suhitha Yerabolu. Co-Supervision with Prof. Riccardo Di Clemente.
- Technical University of Denmark, MSc Human-Centered Artificial Intelligence: Franziska Bäuerlein, Regin Steffansson Kunoy, Diego Rodríguez Gordo, Leonardo Rodovero. Co-Supervision with Prof. Sune Lehmann and Georgios Arvanitidis.

Bachelor Thesis Co-Supervisor

- Scuola Superiore di Catania, Bachelor Degree Physics: Vittorio Turco. Co-Supervision with Prof. Sune Lehmann, Prof. Vito Latora and Prof. Laura Alessandretti

Bachelor Research Project

- Technical University of Denmark, BSc of Artificial Intelligence and Data, Supervising a group of students (4) in their research project about social data science.